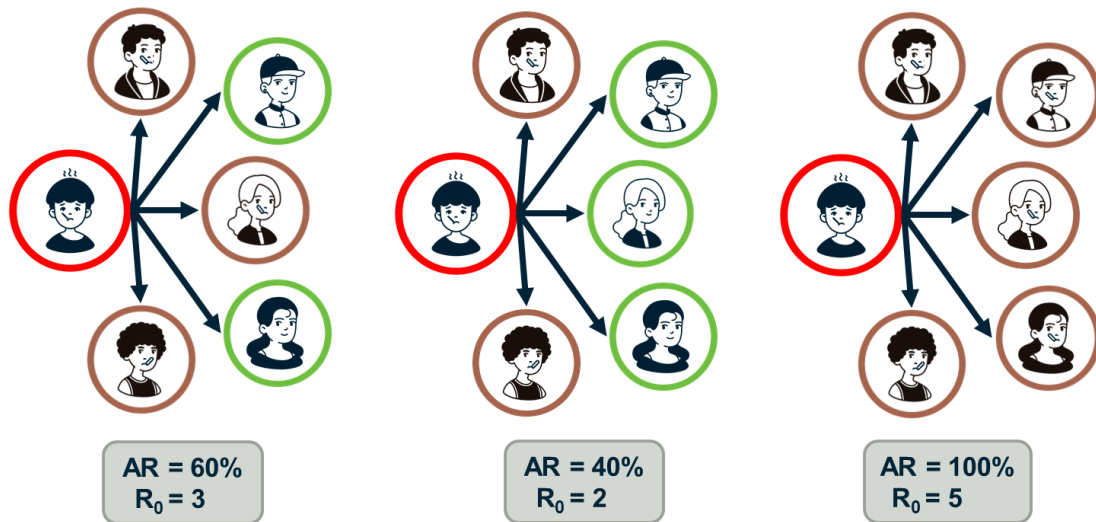


L9: Reproductive Number R_0



Epidemiologists often use the “**reproductive number**”, R_0 , which is *the average number of secondary infections that arise from a single infected individual in a susceptible population*.

One way to estimate R_0 is to multiply the average number of contacts of an infected individual by the probability that a susceptible individual will become infected by a single infected individual (“**attack rate AR**”). So, the R_0 metric does not depend only on the given pathogen – it also depends on the number of contacts each individual has.

If the number of secondary infections from a single infected individual is $R_0 > 1$ then an outbreak is likely to become an epidemic, while if $R_0 < 1$ then an outbreak will not spread beyond a few initially infected individuals.

In the context of the SIS and SIR models, we can easily show that the reproductive number R_0 is equal to the ratio $\frac{\bar{k}\beta}{\mu}$.

The Reproductive Number for Common Infectious Diseases

Disease	Transmission	R_0
Measles	Airborne	12-18
Pertussis	Airborne droplet	12-17
Diphtheria	Saliva	6-7
Smallpox	Social contact	5-7
Polio	Fecal-oral route	5-7
Rubella	Airborne droplet	5-7
Mumps	Airborne droplet	4-7
HIV/AIDS	Sexual contact	2-5
SARS	Airborne droplet	2-5
Influenza (1918 strain)	Airborne droplet	2-3

The table shows the estimated reproductive number for some common infectious diseases.

Note that R_0 also depends on the number of contacts -- and so this metric can vary with time because of interventions such as quarantines, social distancing, or safe-sex practices. The estimates shown in this table should be interpreted as typical values in the absence of such interventions.

Regarding COVID-19, the debate about its actual R_0 is still raging. The first reported result from Wuhan, China was that $R_0=2.2$ -- based on direct contact tracing. As of July 2020, there are estimates in the literature that vary from 2.0 to 6.5.

Food For Thought

Show that in the SIS and SIR models the reproductive number R_0 is equal to the ratio $\frac{\bar{k}\beta}{\mu}$.